

AWS Certified Solutions Architect – Associate

SAA-C03 — THE EXAM BIBLE: FULL EDITION

Everything you must know, condensed. Target score: 950+

Exam Fact	Detail
Questions	65 total (50 scored + 15 unscored), single & multiple response
Time	130 minutes (request +30 min ESL accommodation if English is not your first language)
Passing score	720 / 1000 (scaled). 950+ means missing only ~2–3 scored questions
Cost	\$150 USD
Domain 1	Design Secure Architectures — 30%
Domain 2	Design Resilient Architectures — 26%
Domain 3	Design High-Performing Architectures — 24%
Domain 4	Design Cost-Optimized Architectures — 20%

How to use this bible: Read it 3 times. Pass 1: read everything slowly. Pass 2: focus on every table and EXAM TIP box. Pass 3 (night before exam): read only the Worked Scenarios, Keyword→Answer Map and Exam Traps sections. Pair with timed practice exams — knowledge gets you to 850; pattern recognition from practice gets you to 950+.

0. How the Exam Thinks (Read This First)

Every question is a scenario with constraints. Almost all 4 options are technically possible — you must pick the one matching the **keyword constraint**:

Keyword in question	What AWS wants you to pick
MOST cost-effective / minimize cost	Serverless, Spot, S3 lifecycle, Intelligent-Tiering, single-AZ when DR isn't needed, VPC Gateway Endpoints
LEAST operational overhead / minimal management	Managed & serverless services: Lambda, Fargate, Aurora Serverless, DynamoDB, S3, Athena, EventBridge
Highly available / fault tolerant	Multi-AZ, multiple instances behind ELB + Auto Scaling, Multi-AZ RDS, S3 (already multi-AZ)
MOST performant / lowest latency	Caching (CloudFront, ElastiCache, DAX), read replicas, Global Accelerator, placement groups
MOST secure	Least privilege IAM roles (never keys in code), encryption with KMS, private subnets, VPC endpoints, Secrets Manager
Real-time	Kinesis Data Streams, WebSockets API Gateway, DynamoDB Streams
Near real-time	Kinesis Data Firehose (buffered ~60s)
Decouple	SQS (queue) or SNS (pub/sub) or EventBridge (event bus)

EXAM TIP: Process of elimination wins. First cross out the 1–2 options that violate a hard constraint (wrong service category, wrong latency class, anti-pattern like 'store credentials on the instance'). Then choose between the final 2 using the keyword (cheapest? least overhead? most available?).

- Time math: 130 min / 65 q = 2 min per question. Flag anything taking >3 min and return later.
- Multiple-response questions tell you exactly how many to pick (e.g., 'Choose TWO'). No partial credit.
- 15 questions are unscored experiments — if a question seems insane, answer and move on; it may not count.
- When two answers both work, the more **managed/serverless** one is usually correct unless cost is the keyword.

1. Cloud Foundations (Start Here If You're New)

Concept	Must know
Region	Isolated geographic area (e.g., us-east-1, ca-central-1) with multiple AZs. Choose by: data-residency law, latency to users, service availability, price. Most services are REGIONAL.
Availability Zone (AZ)	One or more physically separate data centers within a region (own power/network), connected by low-latency links. A region has ≥3 AZs. Multi-AZ = survives a data-center failure. Subnets, EC2 instances, and EBS volumes live in ONE AZ.
Edge locations	400+ points of presence worldwide used by CloudFront, Route 53, Global Accelerator — closest to end users.
Global services	IAM, Organizations, Route 53, CloudFront, WAF (with CloudFront). S3 namespace is global but buckets sit in a region. Everything else: assume regional.
Scaling vocabulary	Vertical = bigger instance (has a ceiling, requires downtime). Horizontal = MORE instances (the cloud-native answer; needs a load balancer + stateless app). Elasticity = scale OUT and back IN automatically with demand.
HA vs FT vs DR	High Availability = minimize downtime (Multi-AZ). Fault Tolerance = zero interruption even during failure (more expensive). Disaster Recovery = recover in another region after a major event.
Stateless design	Don't store sessions/files on the instance — push session state to ElastiCache/DynamoDB and files to S3/EFS so any instance can serve any request (prerequisite for Auto Scaling).

Shared Responsibility Model (guaranteed 1–2 questions)

AWS is responsible for (security OF the cloud)	YOU are responsible for (security IN the cloud)
Physical facilities, hardware, global network	Your data, encryption choices (at rest & in transit)
Hypervisor and managed-service infrastructure	IAM users/roles/policies, MFA, key rotation
Patching the infrastructure of MANAGED services (RDS/Lambda/DynamoDB OS & engine)	Patching the GUEST OS on EC2; your application code
AZ/region resilience of the platform	Security Groups, NACLs, firewall config, S3 bucket policies

EXAM TIP: Quick test: if you can SSH into it or configure it, it's yours. 'Who patches the RDS operating system?' = AWS. 'Who patches the OS on EC2?' = You. 'Who configures the security group?' = Always you.

Well-Architected Framework — 6 pillars

Operational Excellence (run & monitor, IaC, small reversible changes) • **Security** (least privilege, encrypt everything, traceability) • **Reliability** (multi-AZ, auto-recovery, test failure) • **Performance Efficiency** (right service/size, serverless, caching) • **Cost Optimization** (pay only for what you use, measure) • **Sustainability** (maximize utilization, managed services). Exam questions are literally written from these pillars — the four domains map to Security, Reliability, Performance, and Cost.

Support plans (occasional question)

Basic = free, no tech support. **Developer** = business-hours email. **Business** = 24/7 phone/chat, FULL Trusted Advisor checks, <1 h response for production down. **Enterprise** = TAM (Technical Account Manager), concierge, <15 min for business-critical down.

2. IAM & Identity (Domain 1 core)

Concept	Must know
Users / Groups	Users = people/apps with long-term credentials. Groups hold users (not other groups). Best practice: permissions via groups.
Roles	Temporary credentials via STS. THE answer for: EC2 accessing S3, cross-account access, Lambda permissions, federation. NEVER hardcode access keys.
Policies	JSON: Effect / Action / Resource / Condition. Explicit Deny ALWAYS wins > explicit Allow > implicit deny (default).
Identity vs Resource policy	Identity-based: attached to user/group/role. Resource-based: attached to resource (S3 bucket policy, SQS policy, KMS key policy) — enables cross-account access without role switching.
Permissions boundary	Max permissions an identity-based policy can grant to a user/role. Does not grant anything itself.
STS	AssumeRole, AssumeRoleWithSAML, AssumeRoleWithWebIdentity (mobile/web via Cognito). Temporary creds 15 min–12 h.
IAM Identity Center	(ex-AWS SSO) One login for multiple accounts + SAML apps. Integrates with AD / external IdP. The modern answer for workforce multi-account access.
Cognito	User Pools = authentication (sign-up/sign-in, JWT) for your app's users. Identity Pools = exchange identity for temporary AWS credentials (e.g., upload to S3 from mobile). 'Hundreds of thousands of users' = Cognito.
AWS Organizations	Multi-account management, consolidated billing (volume discounts shared), OUs. SCPs : guardrails that LIMIT max permissions (never grant); don't affect the management account; apply to all users incl. root in member accounts.
IAM conditions	aws:SourceIp (IP restriction), aws:RequestedRegion, aws:MultiFactorAuthPresent (force MFA), s3:x-amz-server-side-encryption (force encryption).
Access Analyzer	Finds resources shared with external principals (unintended public/cross-account access).

EXAM TIP: Scenario 'application on EC2 needs to access S3/DynamoDB' → ALWAYS 'attach an IAM role to the instance'. Any option with access keys stored on the instance, in code, or in AMI is wrong.

3. Security & Compliance Services

Service	One-line identity (memorize verbatim)
GuardDuty	Intelligent THREAT DETECTION — analyzes CloudTrail, VPC Flow Logs, DNS logs with ML. Detects crypto-mining, compromised credentials.
Inspector	Automated VULNERABILITY SCANNING of EC2 instances, ECR container images, Lambda functions (CVEs, network exposure).
Macie	ML-based discovery of SENSITIVE DATA / PII in S3 buckets.
Detective	INVESTIGATE root cause of findings (graphs from GuardDuty/VPC/CloudTrail data).
Security Hub	AGGREGATES findings from GuardDuty, Inspector, Macie, etc. into one dashboard; compliance checks (CIS).
CloudTrail	AUDIT log of all API calls (who did what, when, from where). 90 days in Event History; create a trail to S3 for longer. Log file integrity validation = digest files.

Service	One-line identity (memorize verbatim)
AWS Config	Records resource CONFIGURATION history; rules check compliance ('all EBS volumes must be encrypted'); can auto-remediate via SSM. Does NOT prevent — it detects.
WAF	Layer 7 firewall on CloudFront / ALB / API Gateway / AppSync. Blocks SQL injection, XSS, geo-blocking, rate-based rules (HTTP flood).
Shield Standard / Advanced	DDoS protection. Standard = free, automatic (L3/L4). Advanced = \$3k/mo, 24/7 DRT team, cost protection, for ELB/CloudFront/Route 53/Global Accelerator.
Network Firewall	Stateful firewall for the whole VPC (IDS/IPS, domain filtering) at the VPC perimeter.
Firewall Manager	Centrally manage WAF/Shield/SG rules across an ORGANIZATION.
Secrets Manager	Stores secrets WITH automatic rotation (native for RDS/Redshift/DocumentDB). Costs money. Keyword: 'rotate credentials'.
SSM Parameter Store	Free config/secret storage (SecureString uses KMS). NO native automatic rotation. Keyword: 'cost-effective config storage'.
ACM	Free public TLS certificates with auto-renewal for ELB, CloudFront, API Gateway. Cannot export public certs; CloudFront certs must be in us-east-1.

EXAM TIP: Memory hook: **GuardDuty** = guard on duty watching for threats. **Inspector** = inspects instances for vulnerabilities. **Macie** = 'My PII'. Security **Hub** = central hub. These four are constantly swapped as distractors.

4. Encryption & KMS

Topic	Must know
KMS keys	AWS managed (aws/s3...; free, auto-rotated yearly, no control) vs Customer managed (\$1/mo, key policies, rotation optional, can disable/delete with 7–30 day wait). Regional by default; multi-Region keys exist.
Envelope encryption	KMS encrypts a data key; the data key encrypts your data (KMS API limit = 4 KB direct encryption).
CloudHSM	Dedicated single-tenant hardware HSM, FIPS 140-2 Level 3, YOU manage keys. Keywords: 'regulatory requirement for dedicated hardware', 'full control of keys'.
S3 encryption	SSE-S3 (AES-256, default, S3-managed keys) SSE-KMS (audit trail in CloudTrail, key policies; watch KMS API quota throttling — use S3 Bucket Keys to reduce calls) SSE-C (customer-supplied key per request, HTTPS only) Client-side.
EBS encryption	KMS-based; encrypts volume, snapshots, and data in transit between instance and volume. Can't directly encrypt an existing unencrypted volume → snapshot → copy snapshot with encryption → create new volume. Enable 'encryption by default' per region.
RDS encryption	Set at creation. Encrypt existing DB: snapshot → copy with encryption → restore. Replicas inherit encryption.
In transit	TLS everywhere; ACM for certs. Site-to-Site VPN = IPsec encrypted. Direct Connect = NOT encrypted by default (run VPN over DX, or MACsec).

5. VPC & Networking (heavily tested)

Component	Must know
VPC / Subnets	VPC = regional, CIDR /16 to /28. Subnet = one AZ. AWS reserves 5 IPs per subnet (.0 network, .1 router, .2 DNS, .3 future, .255 broadcast). Public subnet = route to IGW.
Internet Gateway	1 per VPC; gives public subnets internet. Instances need public/Elastic IP + route 0.0.0.0/0 → IGW.
NAT Gateway	Private subnet OUTBOUND internet. AWS-managed, AZ-specific (deploy 1 per AZ for HA), scales to 100 Gbps, costs per hour + per GB. NAT instance = legacy, self-managed, cheap, disable source/dest check.
Security Groups	STATEFUL (return traffic auto-allowed). Allow rules only. Attached to ENI/instance. Can reference other SGs. Evaluates all rules.
Network ACLs	STATELESS (must allow return traffic on ephemeral ports 1024–65535). Allow AND deny rules, numbered, evaluated in order. Subnet level. Use to BLOCK a specific IP.
VPC Endpoints — Gateway	S3 and DynamoDB ONLY. FREE. Route-table entry. The cost-saving answer for 'private subnet EC2 → S3 without NAT charges'.
VPC Endpoints — Interface	PrivateLink ENI with private IP for most AWS services + partner/custom services. Hourly + per-GB cost. 'Expose a service to other VPCs without peering' = PrivateLink + NLB.

Component	Must know
VPC Peering	1:1 private connection, NOT transitive (A↔B, B↔C does NOT give A↔C), no overlapping CIDRs. Works cross-region/cross-account.
Transit Gateway	Hub-and-spoke router for MANY VPCs + on-prem (VPN/DX). Transitive. The answer for 'hundreds of VPCs' or 'simplify network topology'. Supports multicast.
Site-to-Site VPN	IPsec over internet, 2 tunnels (HA), ~1.25 Gbps per tunnel, minutes to set up. Needs Virtual Private Gateway (AWS side) + Customer Gateway (on-prem side).
Direct Connect	Dedicated private fiber 1/10/100 Gbps. Consistent low latency, weeks-months to provision, not encrypted (add VPN over it). DX + VPN as backup = common HA pattern. DX Gateway = connect DX to multiple VPCs/regions.
VPC Flow Logs	IP traffic metadata (accept/reject) at VPC/subnet/ENI level → CloudWatch Logs or S3. Troubleshoot 'SG/NACL blocking traffic'. Does NOT capture packet contents.
ELB types	ALB = Layer 7 (HTTP/HTTPS, path/host/header routing, Lambda targets, WebSockets). NLB = Layer 4 (TCP/UDP, millions req/s, STATIC IP / Elastic IP per AZ, ultra-low latency). GWLB = Layer 3, deploy 3rd-party firewalls/appliances (GENEVE 6081).
Auto Scaling	Target tracking (keep CPU at 50% — most common answer) Step Scheduled (predictable spikes) Predictive (ML forecast). Cooldown default 300s. Scales across AZs; health checks via ELB recommended. Lifecycle hooks for setup/teardown actions.

EXAM TIP: 'Static IP + millions of TCP requests' = NLB. 'Route /api to one target group and /images to another' = ALB. 'Inspect traffic with 3rd-party appliance' = Gateway Load Balancer. Clients see only ALB's DNS name (no static IP) — if a fixed IP is required for whitelisting, put Global Accelerator in front or use NLB.

6. Route 53, CloudFront, Global Accelerator

Route 53 routing policies

Policy	Use case
Simple	One record → one or more values (random if multiple). No health checks.
Weighted	% traffic split — A/B testing, gradual migration (10% new version).
Latency	Route to region with lowest latency for the user.
Failover	Active-passive DR: primary + health check; secondary (often S3 static site) on failure.
Geolocation	Route by user's LOCATION — content localization, legal/compliance restrictions.
Geoproximity	Route by location with a BIAS to shift traffic between regions (requires Traffic Flow).
Multivalue	Up to 8 healthy records returned — client-side load balancing with health checks.

Alias records: free, point zone apex (example.com) to ELB/CloudFront/S3 website/Global Accelerator. CNAME cannot be used at zone apex. Route 53 = also domain registrar; health checks can monitor endpoints, other health checks (calculated), and CloudWatch alarms.

CloudFront vs Global Accelerator

	CloudFront	Global Accelerator
What	CDN — CACHES content at 400+ edge locations	Routes traffic over AWS backbone, NO caching
Protocols	HTTP/HTTPS	TCP and UDP (gaming, VoIP, IoT)
IPs	Domain name	2 static anycast IPs (whitelisting!)
Best for	Static + dynamic web content, video	Non-HTTP apps, instant regional failover, static IP needs
Origins	S3, ALB, EC2, any HTTP server; OAC restricts S3 to CloudFront only	ALB, NLB, EC2, Elastic IP

EXAM TIP: S3 + CloudFront: block public bucket access and use Origin Access Control (OAC) so content is ONLY reachable via CloudFront. Signed URLs (one file) / signed cookies (many files) for paid private content. Field-Level Encryption protects sensitive POST data. Lambda@Edge / CloudFront Functions = run code at the edge.

7. Compute: EC2, Lambda, Containers

EC2 purchasing options (Domain 4 favorite)

Option	Discount	When
On-Demand	0%	Unpredictable, short-term, can't be interrupted
Reserved (1/3 yr)	Up to 72%	Steady-state known workloads (e.g., DB server). Standard (no type change) vs Convertible (can change family)
Savings Plans	Up to 72%	Commit \$/hr for 1/3 yr. Compute SP = flexible across EC2/Fargate/Lambda, any region. EC2 Instance SP = family+region locked, bigger discount
Spot	Up to 90%	Fault-tolerant, stateless: batch, big data, CI. 2-minute interruption warning. NEVER for databases or critical apps
Dedicated Host	—	Physical server: BYOL licensing (per-socket/per-core), compliance
Dedicated Instance	—	Hardware not shared with other accounts; no host visibility/affinity

EC2 essentials

Topic	Must know
Instance families	C = compute (HPC, gaming), M = general, R = RAM (in-memory DB/cache), T = burstable (credits), I/D = storage IOPS (NoSQL, warehousing), G/P = GPU (ML, video)
Placement groups	Cluster = same rack/AZ, lowest latency, HPC (fails together). Spread = max 7 instances/AZ, critical apps needing isolation. Partition = up to 7 partitions/AZ, Hadoop/Kafka/Cassandra
Metadata	http://169.254.169.254/latest/meta-data — instance info, temp role credentials. IMDSv2 (token-based) is the secure answer
User data	Bootstrap script, runs ONCE at first launch as root
Hibernate	Saves RAM to encrypted EBS — fast resume of long-warm-up apps
AMI	Region-specific (copy across regions); golden AMI = pre-baked for fast Auto Scaling launches

Lambda & serverless limits (memorize numbers)

Limit	Value
Max execution time	15 minutes (longer → Step Functions, ECS/Fargate, or Batch)
Memory	128 MB – 10,240 MB (CPU scales with memory)
Ephemeral /tmp	512 MB default, up to 10 GB
Deployment package	50 MB zipped / 250 MB unzipped; container images up to 10 GB
Concurrency	1,000 default per region (soft limit); reserved concurrency caps a function; provisioned concurrency kills cold starts
VPC access	Attach to VPC for RDS etc.; needs ENI; use RDS Proxy to pool DB connections (Lambda storms exhaust RDS connections)
API Gateway timeout	29 seconds — classic trap: long Lambda behind API GW fails at 29 s even though Lambda allows 15 min

Containers

Service	Identity
ECS	AWS-native container orchestration. Launch types: EC2 (you manage instances) or Fargate (serverless — 'no infrastructure management' answer)
EKS	Managed Kubernetes — keyword: 'Kubernetes', vendor-neutral, hybrid/multi-cloud
Fargate	Serverless compute for ECS/EKS — pay per task, no patching
ECR	Container registry (image scanning, lifecycle policies)
App Runner	PaaS: web app from container/code with zero infra knowledge
Elastic Beanstalk	PaaS for classic apps (Java/.NET/Node...): you keep full control of underlying resources; only the app code is yours to manage. 'Quickly deploy without managing infra but using EC2' = Beanstalk

8. Storage: S3, EBS, EFS, FSx

S3 storage classes (Domain 4 gold)

Class	Min duration	Retrieval	Use case
Standard	—	ms	Frequent access; 11 nines durability, ≥3 AZs (all classes except One Zone)
Intelligent-Tiering	—	ms	UNKNOWN/changing access patterns — auto-moves objects, no retrieval fees, small monitoring fee
Standard-IA	30 days	ms	Infrequent but needs instant access (backups, DR); retrieval fee per GB
One Zone-IA	30 days	ms	Re-creatable data only (1 AZ, 20% cheaper than Standard-IA)
Glacier Instant	90 days	ms	Archive accessed ~once/quarter, instant retrieval
Glacier Flexible	90 days	Expedited 1–5 min / Standard 3–5 h / Bulk 5–12 h	Archives, retrieval in minutes-hours
Glacier Deep Archive	180 days	Standard 12 h / Bulk 48 h	CHEAPEST. Compliance archives kept 7–10 yrs, rare access

S3 features

Feature	Must know
Lifecycle rules	Transition between classes + expire. Min 30 days in Standard before →IA. Clean up incomplete multipart uploads (cost saver)
Versioning	Protects against deletes/overwrites (delete = delete marker). Required for replication. MFA Delete for extra protection
Replication	CRR (cross-region: compliance, latency, DR) / SRR (same-region: log aggregation). Async; only NEW objects (use Batch Replication for existing); versioning required on both
Object Lock	WORM. Compliance mode = nobody (even root) can delete during retention. Governance mode = privileged users can. Legal hold = indefinite. Keyword: 'regulatory, immutable'
Performance	3,500 PUT / 5,500 GET per second PER PREFIX (use many prefixes). Multipart upload >100 MB (required >5 GB). Transfer Acceleration = fast long-distance uploads via edge. Byte-range fetches = parallel downloads
Event notifications	→ SQS / SNS / Lambda / EventBridge (EventBridge = filtering + many destinations)
Access control	Block Public Access (account/bucket level — leave ON), bucket policies (cross-account, force TLS via aws:SecureTransport), presigned URLs (temporary upload/download access)
Static website	Bucket website endpoint + CloudFront; Route 53 alias to apex
Requester Pays	Downloader pays transfer costs (large public datasets)

Block & file storage

Service	Must know
EBS gp3	General SSD. Baseline 3,000 IOPS / 125 MBps regardless of size, scale to 16,000 IOPS independently of size (gp2 ties IOPS to size → gp3 is the cost-optimization answer)
EBS io1/io2	Provisioned IOPS up to 64,000 (256k io2 Block Express). Sub-ms, mission-critical DBs. ONLY type supporting Multi-Attach (same AZ, cluster-aware FS, up to 16 instances)
EBS st1 / sc1	HDD throughput: st1 = big-data/log streaming; sc1 = coldest, cheapest. CANNOT be boot volumes
EBS facts	AZ-locked (move via snapshot). Snapshots = incremental, to S3, copy cross-region/account. Fast Snapshot Restore = no first-read latency (\$\$). Data Lifecycle Manager automates snapshots. DeleteOnTermination: root default true, can change
Instance Store	Physically attached NVMe = highest IOPS (millions). EPHEMERAL — lost on stop/terminate. Buffer/cache/scratch only
EFS	Managed NFS for LINUX, multi-AZ, thousands of concurrent instances, scales to petabytes, pay-per-use (3x gp2 price). Storage classes Standard/IA/Archive + lifecycle. Performance: General Purpose vs Max I/O; Bursting vs Elastic/Provisioned throughput. NOT for Windows
FSx for Windows	Managed Windows file share: SMB, NTFS, Active Directory integration, DFS. Keyword: 'Windows' + 'file share'

Service	Must know
FSx for Lustre	HPC / ML / video processing: sub-ms, 100s GBps. S3 integration (lazy-load & write back). Scratch (temp, cheap) vs Persistent. Keywords: 'HPC', 'machine learning training', 'POSIX + S3'
FSx NetApp ONTAP / OpenZFS	ONTAP: NFS+SMB+iSCSI, snapshots, dedup, migrate from NetApp. OpenZFS: ZFS migrations, NFS
Storage Gateway	Hybrid on-prem↔AWS. S3 File GW : NFS/SMB → S3 with local cache ('extend on-prem storage to S3'). Volume GW : iSCSI; Cached (primary in S3) vs Stored (primary on-prem, async backup). Tape GW : replace physical tape backup with virtual tapes in S3/Glacier
Snow Family	Offline petabyte transfer when network is too slow (rule: >1 week to transfer online → Snowball). Snowball Edge = 80 TB usable + edge compute. Data lands in S3

EXAM TIP: Shared storage decision: Linux many instances = EFS. Windows = FSx for Windows. HPC/ML = FSx for Lustre. Two EC2 in SAME AZ block-level = io2 Multi-Attach. Highest possible local IOPS, data loss OK = Instance Store.

9. Databases

RDS & Aurora

Topic	Must know
RDS engines	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, Db2 (+ RDS Custom for OS access)
Multi-AZ	SYNCHRONOUS standby in another AZ. For AVAILABILITY only — no reads. Auto-failover ~60–120 s (DNS swap, no app change). Multi-AZ DB Cluster variant = 2 READABLE standbys
Read Replicas	ASYNCR. For READ SCALING (up to 15 within region / cross-region / cross-account). Can be promoted (DR). App needs separate read endpoint. Replica of a Multi-AZ DB is allowed; replicas can be Multi-AZ
RDS Proxy	Connection pooling (Lambda!), reduces failover time ~66%, enforces IAM auth
Backups	Automated daily + 5-min transaction logs, retention 0–35 days, point-in-time restore. Manual snapshots = kept forever. Restores create a NEW instance
Aurora	MySQL/PostgreSQL-compatible, 5x/3x faster. 6 copies of data across 3 AZs, self-healing storage auto-grows to 128 TB. Up to 15 replicas w/ reader endpoint + auto-failover <30 s. Backtrack = rewind without restore
Aurora Serverless v2	Auto-scaling capacity (ACUs) — 'intermittent/unpredictable workloads', dev/test
Aurora Global Database	1 primary region + up to 5 read-only secondary regions, replication lag <1 s, promote secondary in <1 min (RPO 1 s, RTO <1 min). THE multi-region RDBMS DR answer

DynamoDB

Topic	Must know
Basics	Serverless NoSQL key-value, single-digit MILLISECOND latency at any scale, multi-AZ by default. Item max 400 KB (bigger → store in S3, pointer in DynamoDB)
Capacity	On-demand (unpredictable, pay-per-request) vs Provisioned RCU/WCU (+auto scaling, cheaper for steady)
DAX	In-memory cache in front of DynamoDB — MICROSECOND reads, no app rewrite. vs ElastiCache: DAX is DynamoDB-specific
Global Tables	Multi-region ACTIVE-ACTIVE replication (needs Streams). Low-latency global apps
Streams + TTL	Streams: item-change events → Lambda (24 h retention). TTL: auto-expire items free (sessions!)
Extras	PITR 35 days; transactions; S3 export; Standard-IA table class for cold tables

Other databases — pick by keyword

Service	Keyword that selects it
ElastiCache Redis	Caching + needs HA/replication/persistence, sorted sets → GAMING LEADERBOARDS, pub/sub, session store
ElastiCache Memcached	Simplest cache, multi-threaded, NO persistence/replication/backup
Redshift	Data WAREHOUSE, OLAP, petabyte analytics, columnar, BI/SQL. Spectrum = query S3 directly. Serverless option

Service	Keyword that selects it
OpenSearch	Full-text SEARCH, log analytics (with Kinesis/CloudWatch), dashboards
Neptune	GRAPH database — social networks, recommendations, fraud rings
DocumentDB	MongoDB-compatible JSON documents
Keyspaces	Apache Cassandra-compatible
Timestream	TIME-SERIES (IoT metrics) — faster/cheaper than relational for it
QLDB	Immutable cryptographically verifiable LEDGER (centralized)
MemoryDB for Redis	Redis-compatible PRIMARY database (durable, multi-AZ) — not just a cache

EXAM TIP: Latency ladder: milliseconds = DynamoDB. Microseconds = DAX or ElastiCache. 'SQL + analytics on huge data' = Redshift (or Athena if data already in S3 and query is ad-hoc/serverless).

10. Decoupling & Application Integration

Service	Must know
SQS Standard	Unlimited throughput, at-least-once delivery, best-effort ordering. Retention 4 days default (max 14). Msg max 256 KB (bigger → Extended Client Library + S3). Visibility timeout 30 s default (max 12 h) — 'message processed twice' = increase it. Long polling (20 s) reduces cost/empty receives. DLQ for poison messages (+redrive). Queue depth (ApproximateNumberOfMessages) drives Auto Scaling of consumers
SQS FIFO	Ordering + EXACTLY-ONCE (dedup). 300 msg/s (3,000 batched, higher in high-throughput mode). Name ends .fifo. MessageGroupId = parallel ordered streams
SNS	Pub/sub PUSH to up to 12M subscribers: email, SMS, HTTP, SQS, Lambda, Kinesis Firehose. FAN-OUT pattern = SNS topic → multiple SQS queues (e.g., order event → billing + shipping + analytics). Message filtering by attributes. FIFO SNS → FIFO SQS
EventBridge	Serverless EVENT BUS: AWS service events, SaaS partners, custom apps. Content-based RULES route to 20+ targets. Scheduler/cron ('run Lambda every night'). Archive & replay. The modern default for event-driven architectures
Kinesis Data Streams	REAL-TIME streaming (sub-second). Shards (1 MB/s in, 2 MB/s out each) or On-demand. Retention 24 h default → 365 days, data REPLAYABLE, multiple consumers. Order per partition key. Consumers: Lambda, KCL apps, Firehose
Kinesis Data Firehose	NEAR-real-time (buffered) delivery, fully managed, ZERO admin: load streams INTO S3, Redshift, OpenSearch, HTTP, Splunk. Optional Lambda transform. No storage/replay. 'Stream logs into S3 with least effort' = Firehose
Managed Flink (ex-Kinesis Analytics)	SQL/Flink analytics ON streams in real time
Amazon MQ	Managed ActiveMQ/RabbitMQ — MIGRATING apps using open protocols (MQTT, AMQP, STOMP, JMS) without rewriting. New apps → SQS/SNS
Step Functions	Visual WORKFLOW orchestration of Lambda/services: sequencing, branching, retries, human approval, parallel. >15 min processes. Standard (1 yr, exactly-once) vs Express (5 min, high-volume)
AppSync	Managed GraphQL (real-time subscriptions, offline sync)
SWF	Legacy workflow — almost always a wrong answer; pick Step Functions

EXAM TIP: SQS vs Kinesis: SQS = decouple work, each message consumed by ONE consumer then deleted, no replay. Kinesis = real-time analytics, MULTIPLE consumers read the same data, replayable, ordered per shard. 'Clickstream/IoT telemetry analytics' = Kinesis. 'Job queue between web tier and workers' = SQS.

11. Analytics & ML Quick Hits

Service	Identity
Athena	Serverless SQL directly on S3. \$5/TB scanned → cut cost with PARTITIONING + columnar formats (Parquet/ORC) + compression. 'Analyze logs in S3 with SQL, least ops' = Athena
Glue	Serverless ETL (Spark) + Data Catalog (central metadata) + Crawlers (auto-discover schemas). Feeds Athena/Redshift/EMR
EMR	Managed Hadoop/Spark clusters — when the question SAYS Hadoop/Spark/big-data frameworks. Use Spot for task nodes
QuickSight	Serverless BI dashboards (pay-per-session)

Service	Identity
Lake Formation	Build/secure a data lake on S3 fast; fine-grained (row/column) access control
MSK	Managed Apache Kafka (keyword: 'Kafka')
Data Exchange	Find/subscribe to 3rd-party data sets
Rekognition	Image/video analysis (faces, objects, content moderation)
Transcribe / Polly	Speech→text / Text→speech
Translate / Comprehend	Language translation / NLP sentiment + entity extraction (PII too)
Textract	Extract text/forms/tables from scanned DOCUMENTS (beyond OCR)
SageMaker	Build/train/deploy custom ML models
Kendra / Personalize / Forecast	Intelligent enterprise search / recommendations / time-series forecasting
Bedrock	Managed generative AI foundation models via API

12. Migration & Transfer

Service	Identity
MGN (Application Migration Service)	Lift-and-shift servers → EC2 (continuous block replication, minimal cutover downtime). Replaces SMS
DMS	Database Migration Service — source DB stays ONLINE during migration. Homogeneous (Oracle→Oracle) or heterogeneous with SCT (Schema Conversion Tool, e.g., Oracle→Aurora). Continuous replication (CDC)
DataSync	ONLINE automated transfer NFS/SMB/HDFS/S3 ↔ S3/EFS/FSx over network (10 Gbps per agent) or DX. Scheduled syncs, preserves metadata. 'Migrate NAS to EFS/S3 over network' = DataSync
Transfer Family	Managed SFTP/FTPS/FTP/AS2 endpoint in front of S3 or EFS — partners keep using their existing SFTP clients
Snow Family	OFFLINE transfer when bandwidth makes online transfer take >~1 week
AppFlow	SaaS (Salesforce, Slack...) → S3/Redshift data flows, no code
Backup	AWS Backup = central, cross-service backup plans (EBS, RDS, DynamoDB, EFS, FSx...), cross-region & cross-account copies, Vault Lock = WORM
7 Rs	Rehost (lift&shift), Replatform (lift-tinker: e.g., DB→RDS), Repurchase (SaaS), Refactor (re-architect cloud-native), Retire, Retain, Relocate (VMware)

13. High Availability & Disaster Recovery

RTO = max acceptable downtime (time to recover). **RPO** = max acceptable data loss (time since last backup/replication). Smaller numbers = more expensive strategy.

Strategy	RTO / RPO	How	Cost
Backup & Restore	Hours / hours	Backups (snapshots, S3, AWS Backup) restored on demand in DR region	\$
Pilot Light	10s of minutes / minutes	Core ONLY always on (e.g., replicated DB); app servers off, started at disaster	\$\$
Warm Standby	Minutes / seconds-min	Full system running SCALED-DOWN in DR region; scale up at disaster	\$\$\$
Multi-site Active-Active	Near zero / near zero	Full capacity in 2+ regions, Route 53 routing both; failover = stop routing to failed region	\$\$\$\$

- Building blocks: Route 53 failover records + health checks, S3 CRR, Aurora Global Database, DynamoDB Global Tables, cross-region read replicas, AMI/snapshot copies, CloudFormation to rebuild fast.
- Within a region, HA = Multi-AZ everything: ELB + ASG across ≥2 AZs (run N+1 so losing an AZ keeps capacity), RDS Multi-AZ, NAT GW per AZ.
- Elastic Disaster Recovery (DRS): continuous block-level replication of on-prem/cloud servers for second-level RPO, minute-level RTO.

EXAM TIP: Match RTO/RPO budget in the question to the strategy. 'RPO of seconds, RTO of minutes, cost-conscious' = Warm Standby. 'Cheapest DR, can tolerate hours' = Backup & Restore. 'Database must survive region failure with <1 s data loss' = Aurora Global.

14. Monitoring, Management & Governance

Service	Identity
CloudWatch Metrics	EC2 default = CPU, network, disk I/O (every 5 min; detailed monitoring = 1 min, \$). MEMORY & disk space need the CloudWatch AGENT (classic trap). Custom metrics; alarms → SNS/Auto Scaling/EC2 actions; composite alarms reduce noise
CloudWatch Logs	Centralize logs (agent, Lambda, VPC Flow); metric filters → alarms ('alert on ERROR in logs'); subscription filters → Kinesis/Lambda; Logs Insights = query language
CloudWatch vs CloudTrail vs Config	CloudWatch = performance (WHAT is happening). CloudTrail = API audit (WHO did it). Config = configuration state & compliance (HOW it changed over time)
EventBridge (ops)	React to state changes: 'EC2 terminated → Lambda', 'GuardDuty finding → SNS'
X-Ray	Distributed TRACING of microservices/Lambda — find the bottleneck/error in a request chain
Trusted Advisor	Account-level checks: cost optimization, security (open SGs, MFA on root), fault tolerance, performance, service quotas. Full checks need Business/Enterprise support
Compute Optimizer	ML rightsizing recommendations for EC2/ASG/EBS/Lambda
Systems Manager (SSM)	Session Manager = shell access with NO SSH/bastion/open ports (audited). Patch Manager, Run Command, Parameter Store, automation runbooks. Agent-based, works hybrid
CloudFormation	Infrastructure as Code — 'reproduce environments / DR rebuild / multi-account consistency (StackSets)'
Service Catalog	Pre-approved CloudFormation products for end users (governance + self-service)
Control Tower	Automated landing zone: multi-account setup with guardrails (uses Organizations)
RAM	Resource Access Manager — share subnets/Transit GW/etc. across accounts without duplication
Health Dashboard	AWS service events affecting YOUR resources (account-specific)
Cost tools	Cost Explorer (analyze + RI/SP recommendations), Budgets (alert/act at thresholds), CUR (most detailed, to S3), Cost Anomaly Detection (ML alerts), cost allocation tags

15. Cost Optimization Patterns (Domain 4)

- **Compute:** Spot for fault-tolerant batch; Savings Plans/RIs for steady state; Graviton (ARM) ~20% cheaper; rightsize with Compute Optimizer; Lambda/Fargate to stop paying for idle; ASG so you don't over-provision.
- **Storage:** gp3 instead of gp2; S3 Intelligent-Tiering for unknown patterns; lifecycle → Glacier; delete unattached EBS volumes & old snapshots; EFS-IA lifecycle; abort incomplete multipart uploads.
- **Data transfer (huge exam theme):** Inbound = FREE. Same-AZ private IP = free. Cross-AZ and cross-region cost money. NAT Gateway processing is pricey → use FREE Gateway VPC Endpoints for S3/DynamoDB. CloudFront egress is cheaper than EC2/S3 egress and caching cuts origin load. Keep chatty services in the same AZ when HA allows.
- **Database:** Aurora Serverless v2 for spiky loads; stop dev RDS instances; read replicas only when needed; DynamoDB on-demand vs provisioned by traffic shape.
- **Analytics:** Athena instead of an always-on cluster; partition + Parquet to slash \$/TB scanned; EMR with Spot task nodes.

EXAM TIP: 'Reduce NAT Gateway charges for S3 traffic' = Gateway VPC Endpoint (free) — one of the most repeated questions on the exam.

16. Reference Architecture Patterns (Memorize These Blueprints)

Most exam scenarios are one of these 8 blueprints with a twist. Learn the chain of services left to right.

Pattern	Blueprint (request flow)
1. Classic 3-tier web app	Route 53 → CloudFront → ALB (public subnets, ≥2 AZs) → Auto Scaling EC2 (private subnets) → RDS Multi-AZ + ElastiCache (private subnets). Static assets in S3 via CloudFront. NAT GW per AZ for outbound. WAF on ALB/CloudFront.
2. Serverless web app	Route 53 → CloudFront → S3 (static SPA, OAC) + API Gateway → Lambda → DynamoDB. Auth = Cognito User Pool. Zero servers, scales to zero cost when idle. The 'least operational overhead' web answer.
3. Event-driven order processing	API/app → SNS topic → fan-out to SQS queues (billing, shipping, analytics) → Lambda or ECS workers → DynamoDB/RDS. DLQs on every queue. Services scale and fail independently.

Pattern	Blueprint (request flow)
4. Streaming analytics / data lake	Producers (apps, IoT) → Kinesis Data Streams → (a) Managed Flink for real-time alerts and (b) Firehose → S3 data lake (Parquet) → Glue Catalog → Athena → QuickSight dashboards.
5. Batch / queue-based processing	Uploads land in S3 → S3 event → SQS → Auto Scaling group of SPOT workers (scales on queue depth) or AWS Batch → results to S3/DynamoDB. Cheap, decoupled, retry-safe.
6. Hybrid enterprise network	On-prem DC ↔ Direct Connect (primary) + Site-to-Site VPN (encrypted backup) → Transit Gateway → many VPCs. Identity: AWS Managed Microsoft AD or AD Connector. Storage bridge: Storage Gateway / DataSync.
7. Multi-region DR (active-passive)	Route 53 failover + health checks. Region A: full stack. Region B: pilot light or warm standby, Aurora Global / DynamoDB Global Tables, S3 CRR, AMIs copied, CloudFormation to scale up on failover.
8. Secure static website	S3 bucket (Block Public Access ON) + CloudFront with OAC + ACM certificate (us-east-1) + Route 53 alias at the apex. Add WAF for protection. Cheapest possible website hosting.

API Gateway essentials (glue for patterns 2–3)

REST API = full features: API keys, usage plans/throttling per client, request validation, caching (0.5–237 GB), WAF. HTTP API = up to 70% cheaper, lower latency, JWT auth — pick for simple Lambda proxies. WebSocket API = two-way real-time (chat). Authorizers: Cognito, Lambda authorizer (custom), IAM. Hard timeout 29 s; payload 10 MB. Edge-optimized (global users via CloudFront) vs Regional vs Private (inside VPC via endpoint).

17. Worked Exam Scenarios — Learn the Elimination Method

Sixteen exam-style questions. For each: cover the answers, pick yours, then study WHY each wrong option is wrong. This is the skill the exam grades.

Q1 (D1 Security). An application on EC2 must read objects from an S3 bucket. What is the MOST secure way to grant access?

- A. Store an IAM user's access keys in the application config — **wrong**: long-term keys on an instance can leak; never the answer
- B. Attach an IAM role to the EC2 instance with s3:GetObject on that bucket — **CORRECT**: temporary, auto-rotated credentials via instance metadata, least privilege
- C. Make the bucket public and restrict by IP — **wrong**: public buckets for app data are an anti-pattern
- D. Embed credentials in the AMI — **wrong**: worse than A — baked into every instance forever

Rule to remember: Compute needs AWS access → IAM ROLE. Any option containing stored access keys is automatically wrong.

Q2 (D1 Security). Security must BLOCK all traffic from a specific malicious IP address to instances in a subnet. What should they use?

- A. Security group deny rule — **wrong**: security groups have NO deny rules — allow only
- B. Network ACL deny rule on the subnet — **CORRECT**: NACLs support explicit DENY, evaluated by rule number, subnet-wide
- C. Remove the IGW — **wrong**: blocks everyone, not just the attacker
- D. Route table blackhole for that IP — **wrong**: route tables direct outbound traffic; not an inbound block mechanism

Rule to remember: DENY a specific IP = NACL (or WAF IP set for HTTP apps). Security groups can only allow.

Q3 (D1 Security). A company must ensure every object uploaded to a bucket is encrypted with SSE-KMS, rejecting non-compliant uploads. How?

- A. Enable default encryption only — **wrong**: encrypts unencrypted uploads but doesn't REJECT uploads that specify a different method; the question demands rejection/enforcement
- B. Bucket policy that denies PutObject unless s3:x-amz-server-side-encryption = aws:kms — **CORRECT**: explicit deny enforces the requirement at the API level
- C. Lambda to scan and re-encrypt nightly — **wrong**: detective + slow, not preventive
- D. Enable versioning — **wrong**: unrelated to encryption

Rule to remember: ENFORCE/PREVENT in S3 = bucket policy with a Deny + condition. Default encryption alone = convenience, not enforcement.

Q4 (D1 Security). An app stores database credentials that must be rotated automatically every 30 days. Which service?

- A. SSM Parameter Store SecureString — **wrong**: encrypts and stores, but NO native automatic rotation
- B. Secrets Manager with rotation enabled — **CORRECT**: built-in scheduled rotation with native RDS integration
- C. KMS — **wrong**: manages encryption KEYS, doesn't store/rotate credentials
- D. Encrypted S3 object — **wrong**: no rotation, awkward retrieval, audit gaps

Rule to remember: The word 'rotate' selects Secrets Manager. 'Cheap/free config storage' selects Parameter Store.

Q5 (D1+D4). EC2 instances in private subnets upload to S3 through a NAT Gateway, causing high data-processing charges. Cheapest fix keeping traffic private?

- A. Gateway VPC Endpoint for S3 + route table entry — **CORRECT**: free, keeps traffic on the AWS network, removes NAT charges for S3
- B. Interface VPC Endpoint for S3 — **wrong**: works but costs hourly + per-GB; question says cheapest
- C. Bigger NAT instance — **wrong**: still pays for processing; self-managed
- D. Move instances to public subnets — **wrong**: violates the security posture

Rule to remember: S3/DynamoDB private access = GATEWAY endpoint (free). Everything else = Interface endpoint (paid).

Q6 (D2 Resilience). A production RDS MySQL database must keep running if its Availability Zone fails, with no manual intervention. What do you enable?

- A. Read replica in another AZ — **wrong**: async + manual promotion = data loss + human action
- B. Multi-AZ deployment — **CORRECT**: synchronous standby, automatic DNS failover in ~1–2 minutes
- C. Daily snapshots to S3 — **wrong**: recovery takes hours and loses up to a day of data
- D. Second DB with application dual-writes — **wrong**: complex, error-prone anti-pattern

Rule to remember: Survive AZ failure automatically = Multi-AZ. Scale reads = read replicas. Don't swap them.

Q7 (D2 Resilience). A web tier sends image-processing jobs to a fleet of EC2 workers. During spikes, jobs are lost. Fix with minimal change?

- A. SQS queue between tiers; workers poll; Auto Scaling on queue depth — **CORRECT**: queue buffers spikes, retries via visibility timeout, scales workers with backlog
- B. Bigger worker instances — **wrong**: vertical scaling has a ceiling and still drops jobs at the limit
- C. Kinesis Data Streams — **wrong**: for streaming analytics with multiple consumers/replay; overkill for a job queue
- D. Direct synchronous calls with retries — **wrong**: keeps tiers coupled; retries amplify load during spikes

Rule to remember: 'Jobs lost during spikes' = put SQS in the middle and scale consumers on ApproximateNumberOfMessages.

Q8 (D2 Resilience). A global app needs its relational database to survive a REGION failure with RPO of ~1 second and RTO under 1 minute. Choose:

- A. RDS Multi-AZ — **wrong**: Multi-AZ protects within ONE region only
- B. Cross-region read replica — **wrong**: promotion is manual and lag is unmanaged; RTO/RPO not guaranteed
- C. Aurora Global Database — **CORRECT**: storage-level replication <1 s lag, secondary promoted in <1 min
- D. Nightly snapshot copies to another region — **wrong**: RPO measured in hours

Rule to remember: Region-level RDBMS DR with second-level RPO = Aurora Global Database. (NoSQL equivalent = DynamoDB Global Tables.)

Q9 (D2 Resilience). Private-subnet instances in 3 AZs reach the internet through ONE NAT Gateway. An AZ outage took the app offline. Fix?

- A. One NAT Gateway per AZ; each private subnet routes to its local NAT GW — **CORRECT**: removes the single point of failure and avoids cross-AZ data charges
- B. Second NAT GW in the same AZ — **wrong**: same AZ = same failure domain
- C. Replace with a NAT instance — **wrong**: less available than a NAT GW, self-managed
- D. Add an IGW to private subnets — **wrong**: that makes them public subnets

Rule to remember: NAT Gateway is AZ-scoped. HA = one per AZ + per-AZ route tables.

Q10 (D3 Performance). Customers must whitelist TWO static IPs to reach an application behind an ALB in two regions, with fast regional failover. Choose:

- A. Elastic IPs on the ALB — **wrong**: ALBs do not support Elastic/static IPs
- B. AWS Global Accelerator in front of the ALBs — **CORRECT**: 2 static anycast IPs, AWS backbone routing, near-instant cross-region failover
- C. CloudFront — **wrong**: no static IPs and it's a cache, not an IP front door
- D. Route 53 latency records — **wrong**: DNS gives names, not static IPs; failover delayed by TTL caching

Rule to remember: 'Static IP' + ALB or multi-region or UDP = Global Accelerator. (Single-region TCP only? NLB also works.)

Q11 (D3 Performance). A read-heavy MySQL RDS database is overloaded by repeated identical product-catalog queries. Improve performance MOST effectively:

- A. ElastiCache (Redis) for hot query results + read replicas for remaining reads — **CORRECT**: cache absorbs repeats at microsecond latency; replicas scale what's left
- B. Multi-AZ — **wrong**: availability, zero read scaling on the classic standby
- C. Bigger instance — **wrong**: helps briefly, costly, doesn't remove repeated query load
- D. Move to DynamoDB — **wrong**: re-architecting a relational app isn't a performance quick-win

Rule to remember: Read-heavy + repeated queries = CACHE first, read replicas second. 'Same data requested often' screams ElastiCache.

Q12 (D3 Performance). A genomics workload on hundreds of Linux EC2 instances needs a shared high-performance POSIX file system that loads datasets from S3. Choose:

- A. EFS — **wrong**: general-purpose NFS; not built for HPC throughput or S3-linked lazy loading
- B. FSx for Lustre linked to the S3 bucket — **CORRECT**: purpose-built for HPC/ML: hundreds of GB/s, sub-ms, native S3 integration
- C. FSx for Windows — **wrong**: SMB/Windows — wrong OS world
- D. EBS io2 Multi-Attach — **wrong**: single AZ, max 16 instances, block not shared-file semantics

Rule to remember: HPC / ML training / 'process S3 data with POSIX' = FSx for Lustre. Every time.

Q13 (D3 Performance). Clickstream events must be ingested in real time and consumed simultaneously by a fraud detector AND a dashboard, with replay for 7 days. Choose:

- A. SQS Standard — **wrong**: one consumer deletes the message; no replay; no multi-reader fan-in
- B. Kinesis Data Streams with 7-day retention — **CORRECT**: ordered, replayable stream read by multiple consumers in parallel
- C. Kinesis Data Firehose — **wrong**: delivery-only into S3/Redshift; no replay, no custom real-time consumers
- D. SNS — **wrong**: push fan-out but no retention/replay or ordered stream processing

Rule to remember: Multiple consumers + replay + real-time = Kinesis Data Streams. Just landing data into S3 = Firehose.

Q14 (D4 Cost). Analysts run ad-hoc Athena SQL on 5 TB of JSON logs in S3 and costs are exploding. Reduce query cost the MOST:

- A. Convert to Parquet, partition by date, compress — **CORRECT**: columnar + partition pruning can cut data scanned (the \$5/TB driver) by 90%+
- B. Move logs to Redshift — **wrong**: adds an always-on warehouse cost for ad-hoc queries
- C. Reserved capacity for Athena — **wrong**: doesn't reduce data scanned; wrong lever for ad-hoc
- D. S3 Standard-IA — **wrong**: cuts storage cost, not query cost; adds retrieval fees

Rule to remember: Athena bills per TB SCANNED → scan less: Parquet/ORC + partitions + compression.

Q15 (D4 Cost). A bucket holds millions of objects with UNPREDICTABLE access patterns. Minimize storage cost without performance impact or operational effort:

- A. Lifecycle rule to Standard-IA after 30 days — **wrong**: unpredictable access means IA retrieval fees may exceed savings
- B. S3 Intelligent-Tiering — **CORRECT**: auto-moves each object between tiers by observed access; no retrieval fees, no management
- C. Glacier Flexible after 60 days — **wrong**: minutes-to-hours retrieval breaks 'no performance impact'
- D. One Zone-IA — **wrong**: lower durability scope; still wrong for unknown access

Rule to remember: The literal word 'unpredictable/unknown access pattern' = Intelligent-Tiering. Known cold schedule = lifecycle rules.

Q16 (D4 Cost). A nightly video-transcoding batch job is fault-tolerant and can be re-run if interrupted. It runs 4 hours on 50 instances. Cheapest compute:

- A. On-Demand — **wrong**: pays full price for interruption-tolerant work
- B. Reserved Instances — **wrong**: commits 24/7 pricing for a 4-hour nightly job
- C. Spot Instances — **CORRECT**: up to 90% off; interruption tolerance is exactly the Spot use case
- D. Dedicated Hosts — **wrong**: most expensive; for licensing/compliance only

Rule to remember: Fault-tolerant + restartable + batch = Spot. Steady 24/7 = Savings Plans/RI. Unpredictable + critical = On-Demand.

18. The Keyword → Answer Map (Night-Before Review)

If you see the phrase on the left, the answer is almost always the service on the right.

If the question says...	The answer is...
NoSQL, single-digit millisecond, serverless, any scale	DynamoDB
Microsecond latency cache for DynamoDB	DAX
Caching / session store / leaderboard (sorted sets)	ElastiCache Redis
SQL data warehouse / OLAP / BI on petabytes	Redshift
Ad-hoc SQL directly on files in S3, serverless	Athena (Parquet + partitions to cut cost)
Serverless ETL / data catalog / crawlers	Glue
Hadoop / Spark (named explicitly)	EMR
Graph relationships / fraud ring / social network	Neptune
Time-series IoT sensor data	Timestream
MongoDB-compatible	DocumentDB
Immutable verifiable ledger	QLDB
Full-text search / log analytics + dashboards	OpenSearch
Windows file share / SMB / Active Directory	FSx for Windows File Server
HPC, ML training, parallel file system, S3-linked	FSx for Lustre

If the question says...	The answer is...
Shared POSIX/NFS storage for many Linux EC2	EFS
Block storage shared by instances in ONE AZ	EBS io2 Multi-Attach
Highest IOPS, temporary data acceptable	Instance Store
Backup on-prem tapes to cloud	Storage Gateway — Tape Gateway
On-prem NFS/SMB extending to S3 with local cache	Storage Gateway — S3 File Gateway
Petabytes, limited bandwidth, one-time offline move	Snowball Edge
Ongoing online file migration NFS→AWS, scheduled	DataSync
Partners upload via SFTP	Transfer Family
Migrate database with minimal downtime	DMS (+ SCT if changing engines)
Lift-and-shift servers to EC2	MGN
Decouple producer from consumer / buffer load	SQS
One event → many consumers (fan-out)	SNS → multiple SQS
Strict ordering + exactly-once	SQS FIFO
Real-time streaming, replay, multiple consumers	Kinesis Data Streams
Load streaming data into S3/Redshift/OpenSearch, no admin	Kinesis Data Firehose
Migrate ActiveMQ/RabbitMQ (MQTT/AMQP) without rewrite	Amazon MQ
Orchestrate multi-step workflow, retries, human approval	Step Functions
Cron / schedule / react to AWS events / SaaS events	EventBridge
Static IP for global app / non-HTTP (TCP-UDP) acceleration	Global Accelerator
Cache static+dynamic content globally / protect S3 origin	CloudFront (+ OAC)
Block SQL injection / XSS / rate-limit HTTP	WAF
DDoS protection with response team / cost protection	Shield Advanced
Detect compromised instance / crypto mining	GuardDuty
Scan EC2/containers/Lambda for CVEs	Inspector
Find PII in S3	Macie
Who made this API call?	CloudTrail
Track config changes / enforce 'all volumes encrypted'	AWS Config
Rotate database credentials automatically	Secrets Manager
Dedicated hardware key storage, FIPS 140-2 L3	CloudHSM
Shell access without SSH keys or bastion	SSM Session Manager
Restrict permissions across all accounts in an org	Organizations SCP
Multi-account single sign-on for workforce	IAM Identity Center
Mobile/web app user sign-up & sign-in	Cognito User Pools
App users get temporary AWS credentials	Cognito Identity Pools
Connect hundreds of VPCs + on-prem, transitive	Transit Gateway
Private access to my service from other VPCs, no peering	PrivateLink (Interface Endpoint + NLB)
Private subnet to S3/DynamoDB without NAT cost	Gateway VPC Endpoint
Consistent dedicated network to on-prem	Direct Connect (+ VPN for encryption/backup)
Quick encrypted tunnel to on-prem over internet	Site-to-Site VPN
Kubernetes (the word appears)	EKS
Containers, no infrastructure management	Fargate
Deploy app fast, keep control of resources, classic stack	Elastic Beanstalk
Code runs <15 min, event-driven, no servers	Lambda
WORM / immutable / regulatory retention in S3	S3 Object Lock (Compliance mode)

If the question says...	The answer is...
Unknown/changing S3 access pattern, optimize cost	S3 Intelligent-Tiering
Cheapest archive, retrieval in 12–48 h fine	S3 Glacier Deep Archive
Global RDBMS DR, RPO ~1 s, RTO <1 min	Aurora Global Database
Multi-region active-active NoSQL	DynamoDB Global Tables
Too many DB connections from Lambda	RDS Proxy
Read-heavy relational workload	Read Replicas (+ ElastiCache)
DB must survive AZ failure, no manual action	RDS Multi-AZ

19. Classic Exam Traps (Don't Fall For These)

- **Multi-AZ vs Read Replicas:** Multi-AZ = availability/failover (sync, standby NOT readable on classic Multi-AZ). Read replicas = performance/read scaling (async). If question says 'improve read performance', Multi-AZ is wrong.
- **SG vs NACL:** need to BLOCK/deny a specific IP → NACL (SGs can't deny). Stateful vs stateless return-traffic logic.
- **CloudFront vs Global Accelerator:** caching HTTP content vs accelerating TCP/UDP with static IPs. 'Static IP' or 'UDP' → Global Accelerator.
- **CloudWatch vs CloudTrail:** metrics/performance vs API audit. 'Who deleted the bucket?' = CloudTrail.
- **Memory utilization on EC2:** NOT a default metric — needs CloudWatch Agent (custom metric).
- **API Gateway 29 s timeout** even though Lambda runs 15 min. Long jobs: async (SQS + worker) or Step Functions.
- **SQS message processed twice:** increase VISIBILITY TIMEOUT (not retention, not long polling). Long polling fixes empty-receive cost, not duplicates.
- **Kinesis shard limits / ProvisionedThroughputExceeded:** more shards, better partition key distribution, or on-demand mode.
- **S3 encryption enforcement:** bucket policy denying uploads without the encryption header; force TLS with `aws:SecureTransport=false` → Deny.
- **Existing unencrypted EBS/RDS:** can't encrypt in place — snapshot → encrypted copy → restore/new volume.
- **VPC peering is NOT transitive** and CIDRs must not overlap. Many VPCs → Transit Gateway.
- **SCPs never GRANT permissions** — they cap them. The user still needs an IAM policy. SCPs don't bind the management account.
- **Spot for databases / critical single instances** = always wrong. Spot is for interruptible work only.
- **Lambda >15 min, >10 GB RAM, or needing GPUs** → wrong tool; use Fargate/ECS/Batch.
- **S3 'lost data' protection:** versioning + MFA Delete (accidental) vs Object Lock (compliance/malicious).
- **Glacier retrieval times matter:** 'within minutes' kills Deep Archive (12 h min). 'Instant access archive' = Glacier Instant Retrieval.
- **Aurora reader endpoint vs cluster (writer) endpoint:** send reads to the reader endpoint; don't point apps at instance endpoints.
- **EFS doesn't support Windows;** FSx for Windows doesn't fit Linux POSIX HPC — that's Lustre.
- **NAT Gateway lives in a PUBLIC subnet** with an Elastic IP; private subnets route to it. One per AZ for HA.
- **'Least operational overhead' eliminates** any answer with 'install on EC2', 'cron job on an instance', 'manage a cluster' when a managed equivalent exists.
- **ALB has no static IP.** Need fixed IPs? NLB (or Global Accelerator in front of ALB).
- **Route 53 zone apex (example.com)** → ALIAS record, never CNAME.
- **Instance role credentials** come from instance metadata; option saying 'copy credentials to the instance' is wrong.
- **S3 Transfer Acceleration** = fast UPLOADS over long distance; CloudFront = fast DOWNLOADS/distribution.

20. Final Week Checklist

- Day -7 to -3: one timed practice exam per day (Tutorials Dojo / official). Target ≥90% consistently. Review EVERY wrong AND lucky-guess answer; log the rule you missed.
- Day -2: re-read sections 0, 17, 18, 19 of this bible. Re-do only your wrong-answer log.
- Day -1: light review of the Keyword Map only. Sleep 8 hours — worth more than 3 hours of cramming.
- Exam day: 2 min/question budget; flag and move on; eliminate 2, then apply the keyword (cheapest / least overhead / most available / fastest); never leave blanks — no penalty for guessing.
- Online proctored: clean desk, no second monitor, ID ready, stable internet, bathroom before (no breaks allowed on Pearson VUE online).
- If English is not your first language: request the ESL +30 minutes accommodation BEFORE scheduling.

Reality check for 950+: this bible gives you the knowledge layer. The last 100 points come from doing 300+ practice questions and reading the explanations for every single one. Read this document, drill practice exams, return to sections 17–19 nightly. You've got this.